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NLP & Text Analysis **(COMM040)**

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Intended Learning Outcomes

- Understand the strengths, weaknesses and biases of different NLP methods
- Know about recent advances and applications of NLP methods, their impact and real world applications
- Use modern NLP techniques to derive insight from unstructured data

Outline

- Language Models
- N-gram LM
- Neural LM
- Attention
- Transformer LM

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Language Models

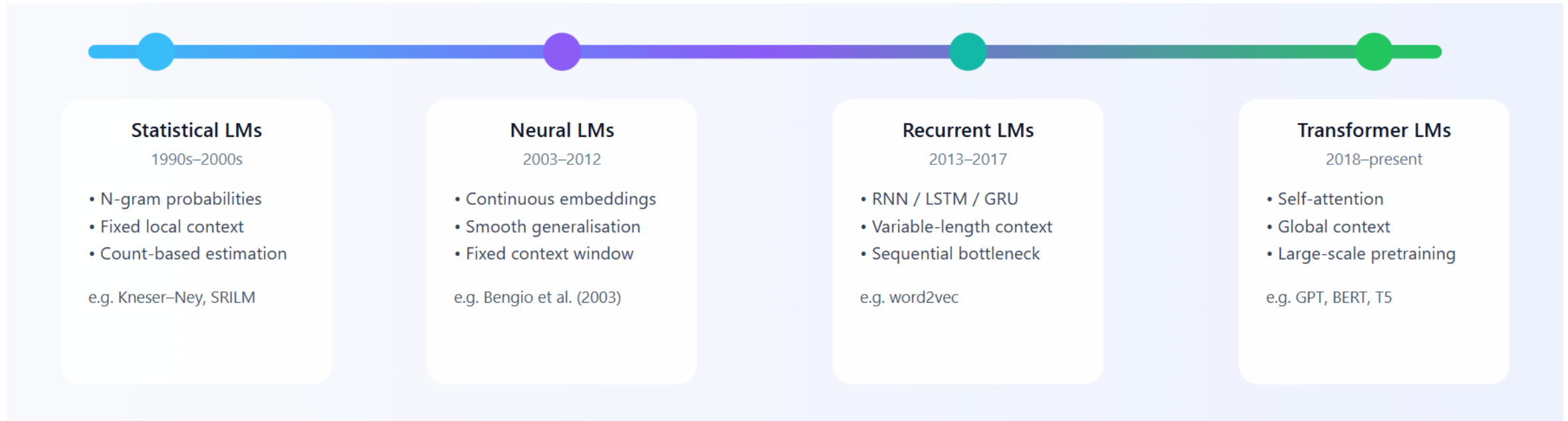
- Predicting upcoming words

The Shannon Game: **How well can we predict the next word?**

- Once upon a _____
- That is a picture of a _____
- For breakfast I ate my usual _____

menti.com

Language Models



Outline

- Language Models
- **N-gram LM**
- Neural LM
- Attention
- Transformer LM

N-gram LM

- n-gram:
 - Goal: compute the probability of a sentence or sequence of words W :

$$P(W) = P(w_1, w_2, w_3, w_4, w_5 \dots w_n)$$

- Related task: probability of an upcoming word:

$$P(w_5 | w_1, w_2, w_3, w_4) \text{ or } P(w_n | w_1, w_2 \dots w_{n-1})$$

N-gram LM

- n-gram:

<s> I am Sam </s>

<s> Sam I am </s>

<s> I do not like green eggs and ham </s>

$$P(\text{I} \mid \text{<s>}) = \frac{2}{3} = .67 \qquad P(\text{Sam} \mid \text{<s>}) = \frac{1}{3} = .33 \qquad P(\text{am} \mid \text{I}) = \frac{2}{3} = .67$$

$$P(\text{</s>} \mid \text{Sam}) = \frac{1}{2} = 0.5 \qquad P(\text{Sam} \mid \text{am}) = \frac{1}{2} = .5 \qquad P(\text{do} \mid \text{I}) = \frac{1}{3} = .33$$

N-gram LM

- n-gram:
 - Probability depends on the size of the test set
Probability gets smaller the longer the text
 - **Perplexity** is the inverse probability of the test set, normalized by the number of words

$$PP(W) = P(w_1 w_2 \dots w_N)^{-\frac{1}{N}}$$

$$= \sqrt[N]{\frac{1}{P(w_1 w_2 \dots w_N)}}$$

N-gram LM

- Describe the model properties:
 - [menti.com](https://www.menti.com)

N-gram LM

- n-gram tools:

- SRILM

- <http://www.speech.sri.com/projects/srilm/>

- KenLM

- <https://kheafield.com/code/kenlm/>

N-gram LM

- n-gram LM for text generation:

Choose a random bigram ($\langle s \rangle$, w) $\langle s \rangle$ I

according to its probability $p(w|\langle s \rangle)$ I want

Now choose a random bigram (w , x) want to
according to its probability $p(x|w)$ to eat

And so on until we choose $\langle /s \rangle$ eat Chinese

Then string the words together Chinese food
food $\langle /s \rangle$

I want to eat Chinese food

N-gram LM

- n-gram:
- can't handle long-distance dependencies:

“**The soups** that I made from that new cookbook I bought yesterday **were** amazingly delicious.”
- don't do well at modeling new sequences with similar meanings

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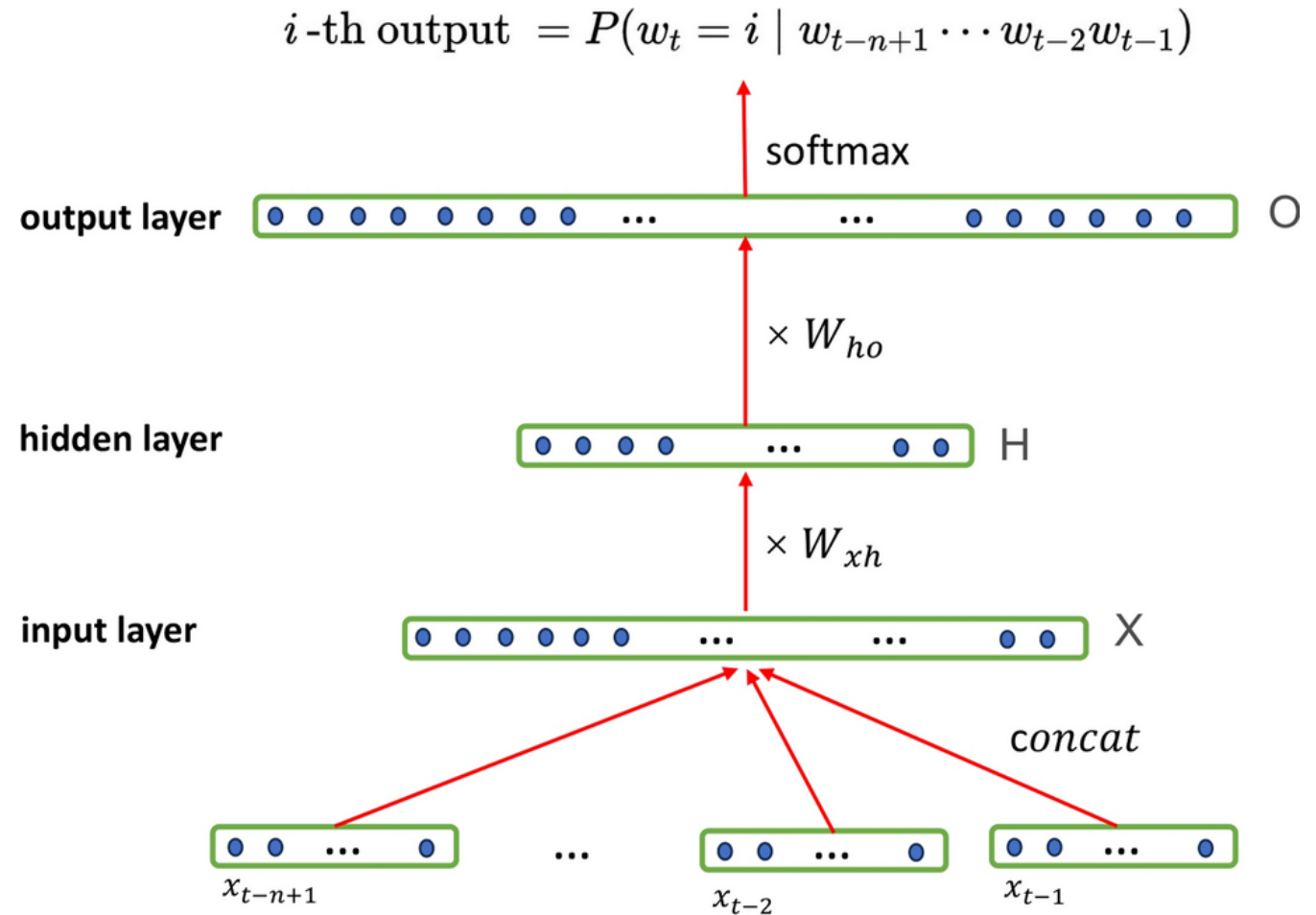
Neural LM

- Idea 1: Defining meaning by linguistic distribution
“Words with similar meanings tend to appear in similar linguistic contexts”
- Idea 2: Meaning as a point in multidimensional space



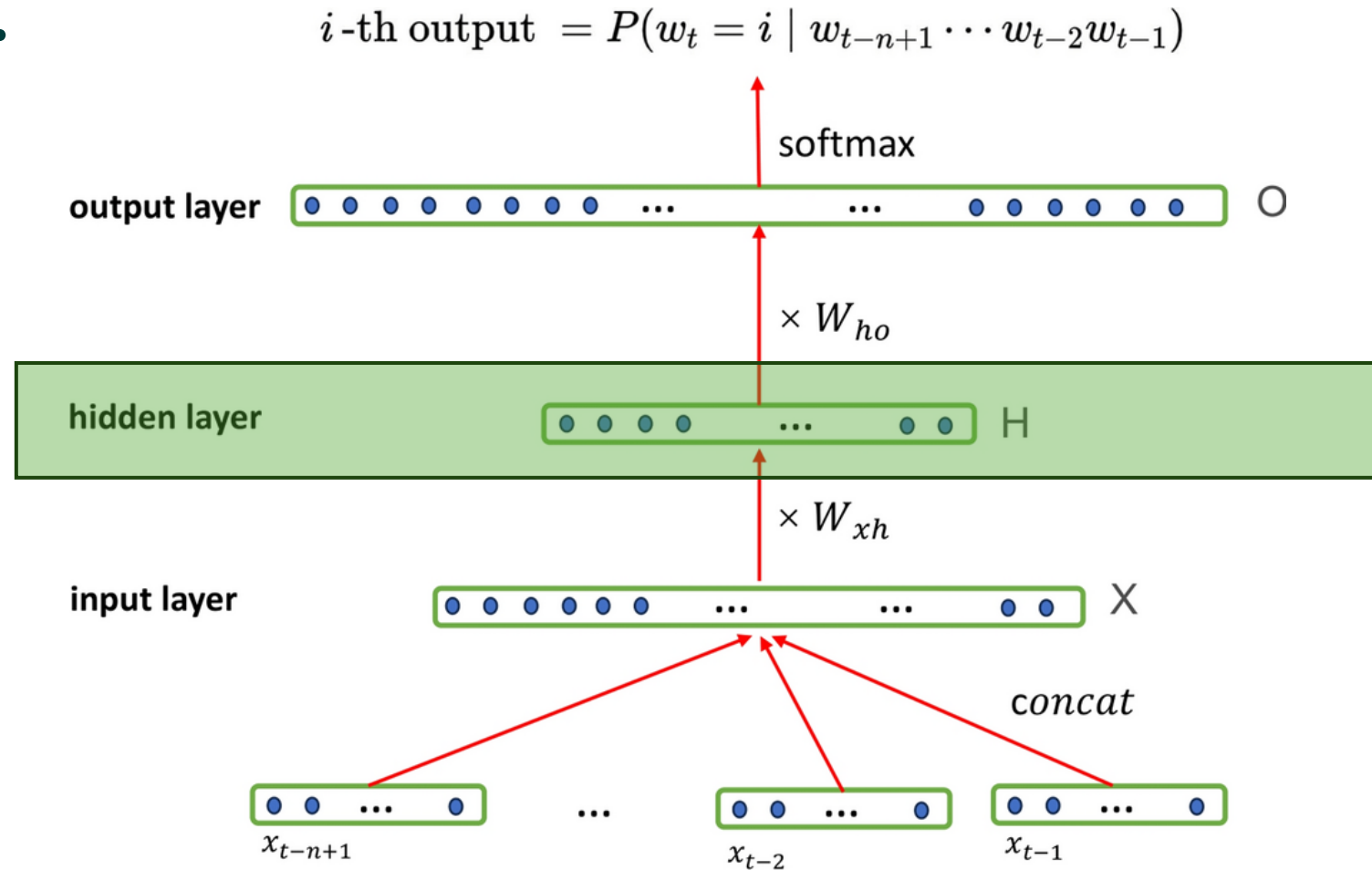
Neural LM

- Word2vec, Glove...



Neural LM

- Word2vec, Glove...



Neural LM

- Word Embedding
 - They are **static**!
 - The embedding for a word doesn't reflect how its meaning changes in context.
 - Stopwords group meanings
 - Very ambiguous words

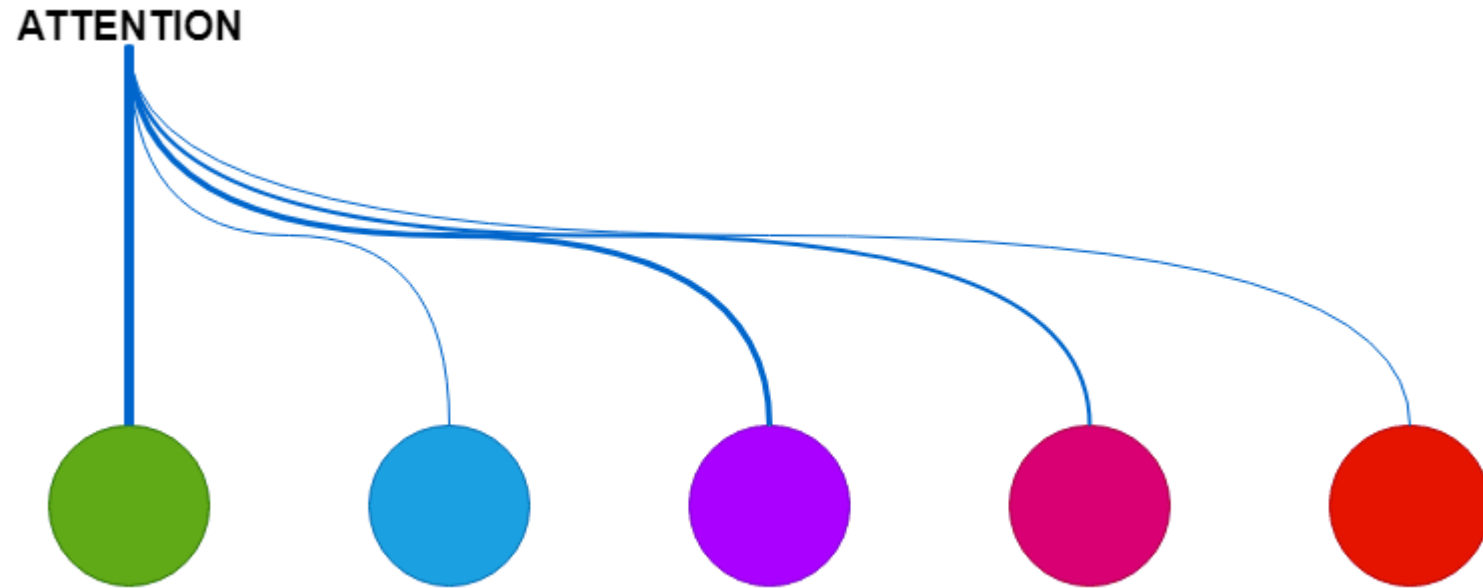
The chicken didn't cross the road because **it** was too tired

- How can two vectors be compared?
 - [menti.com](https://www.menti.com)

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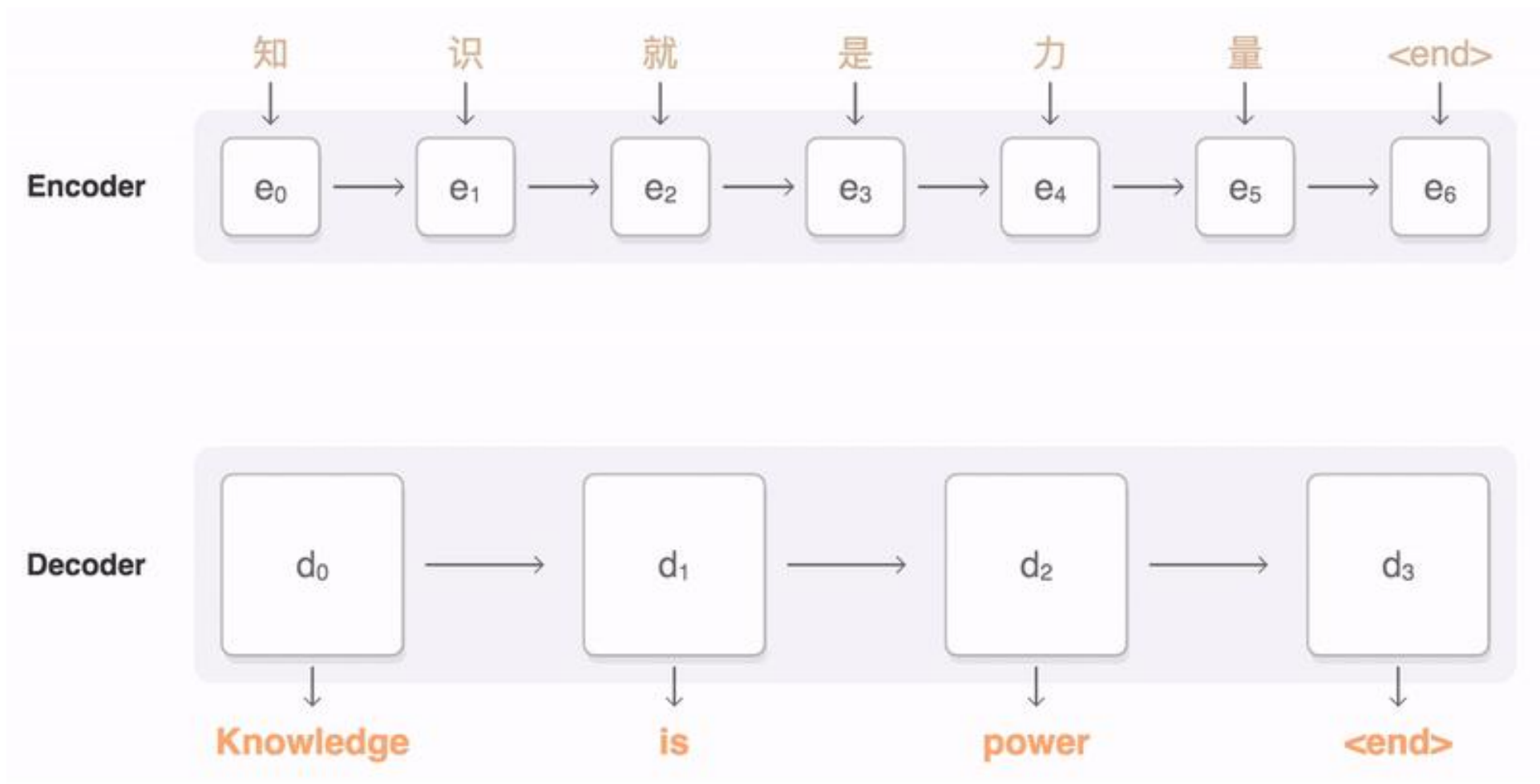
Attention



Attention



Attention



Attention

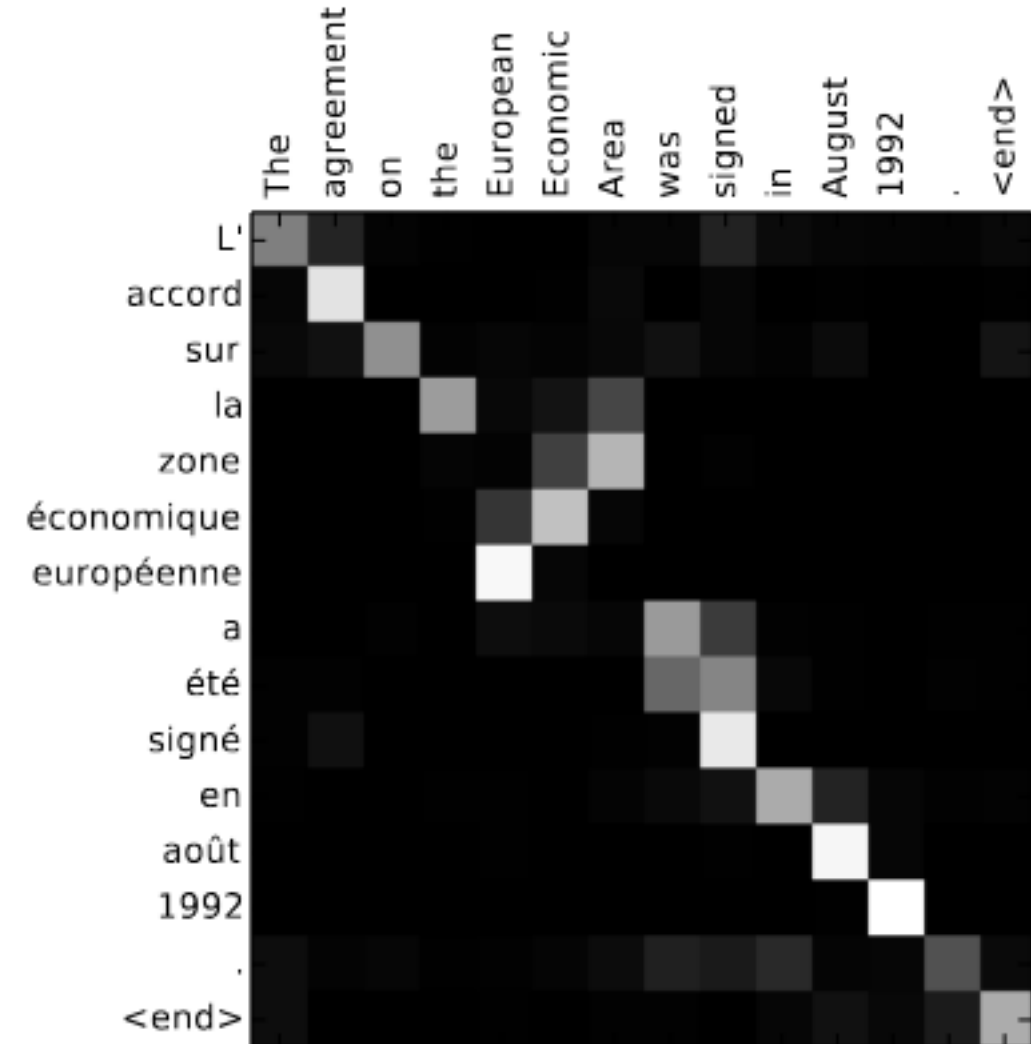


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$$\text{Attention}(q, k, v) = \overbrace{\text{softmax}\left(\frac{qk^T}{\sqrt{d_k}}\right)}^{\text{Attention weights}} v$$

from to

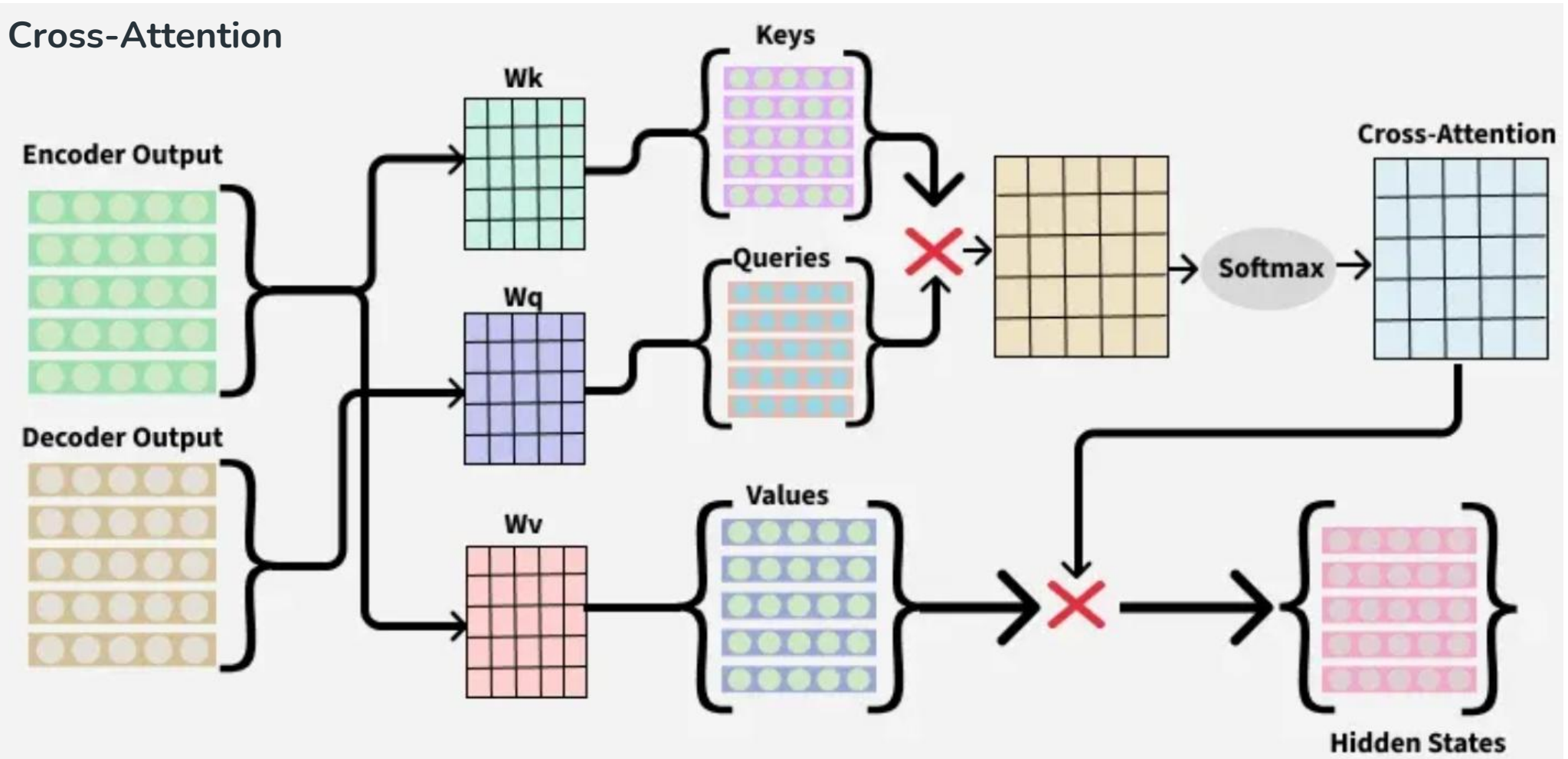
vector dimensionality of K, V



Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation jointly learning to align and translate. *arXiv preprint arXiv:1409.0473*.

Attention

Cross-Attention

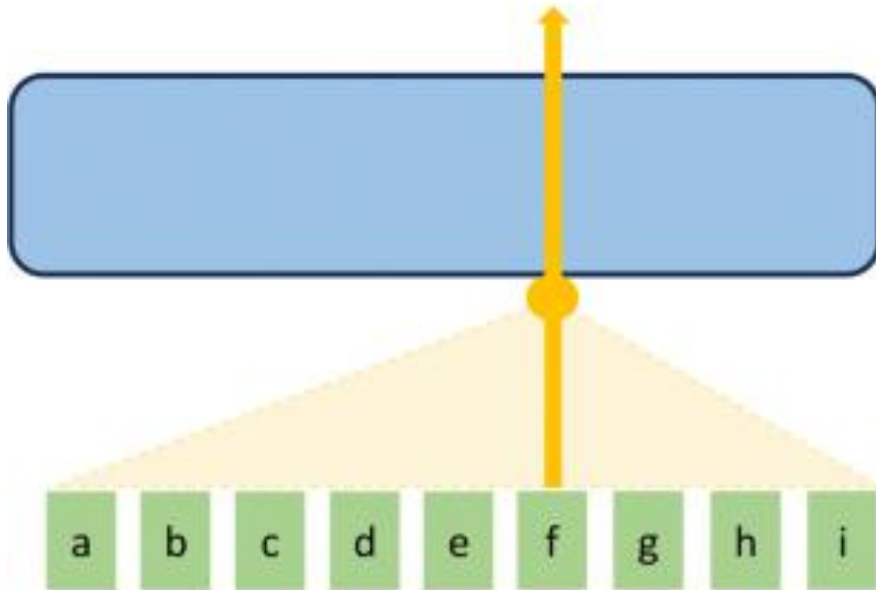


Attention

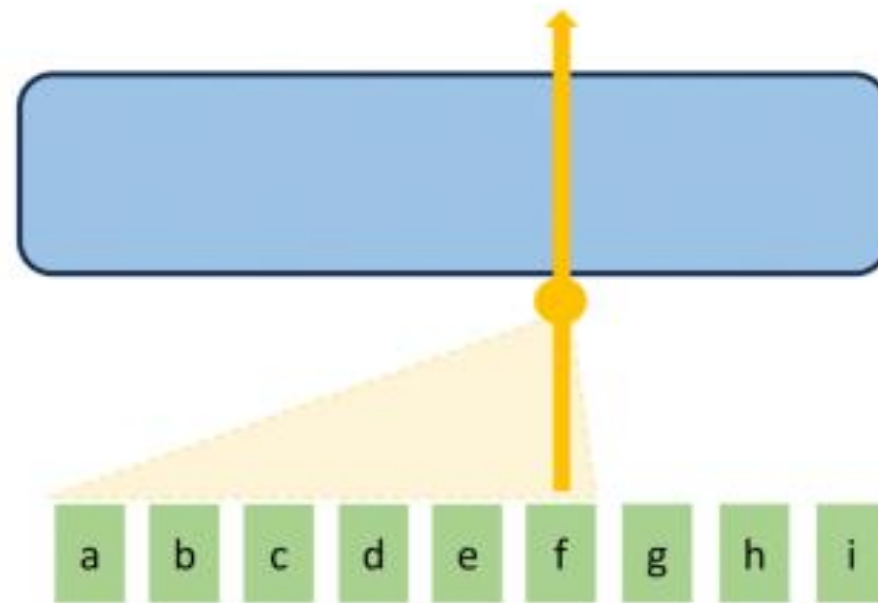
- In the attention mechanism, what does the model compute?
 - [menti.com](https://www.menti.com)

Flavours of self-attention

Self-Attention



Masked Self-Attention



Outline

- Language Models
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- **Transformer LM**

Transformer LM



Attention Is All You Need

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Transformer LM

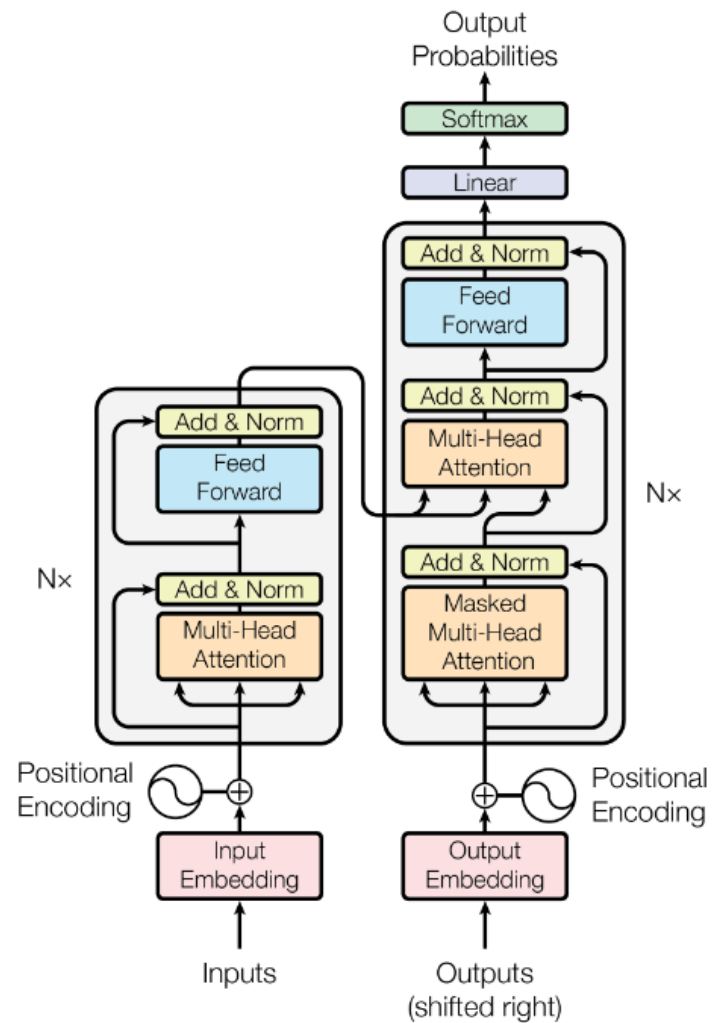
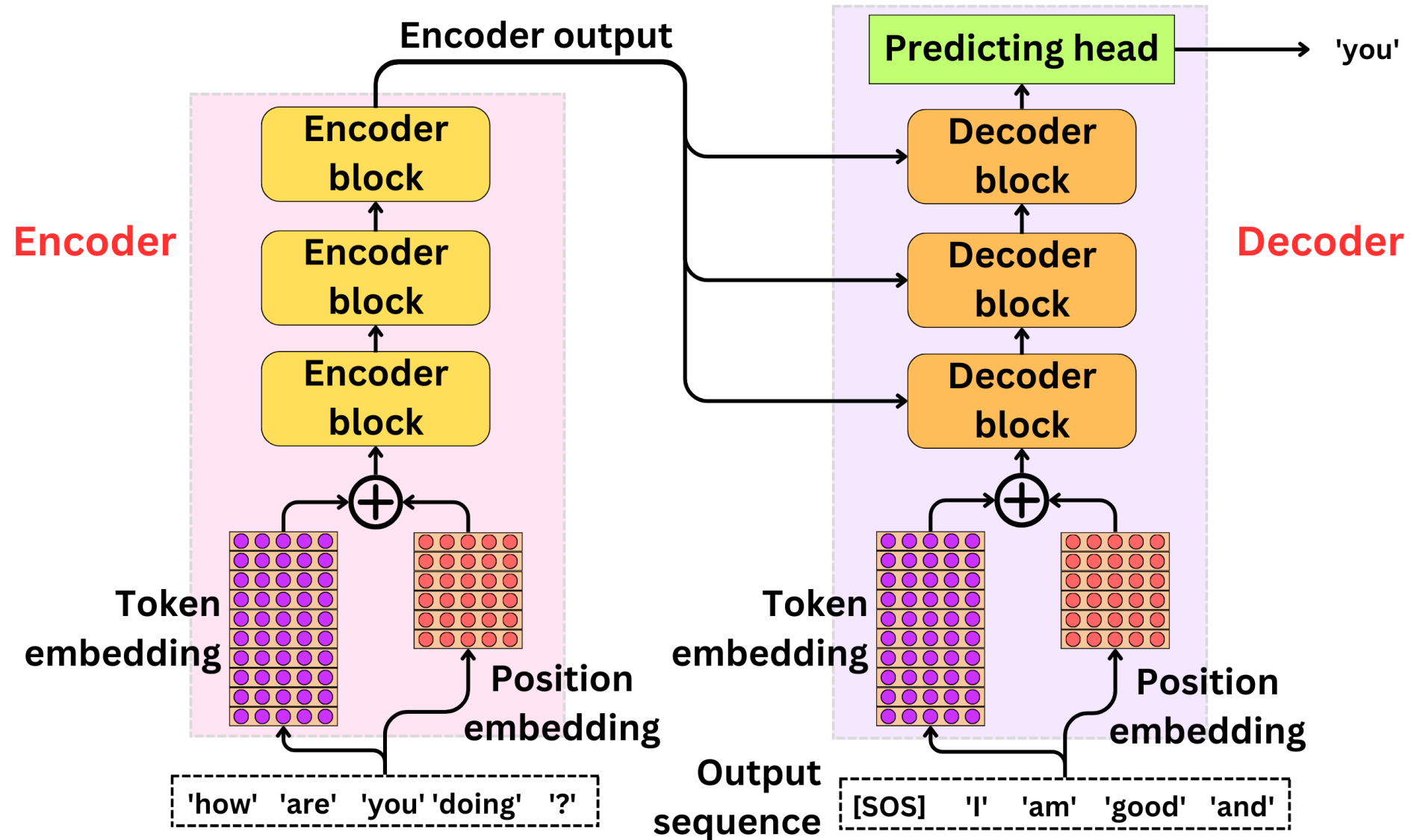
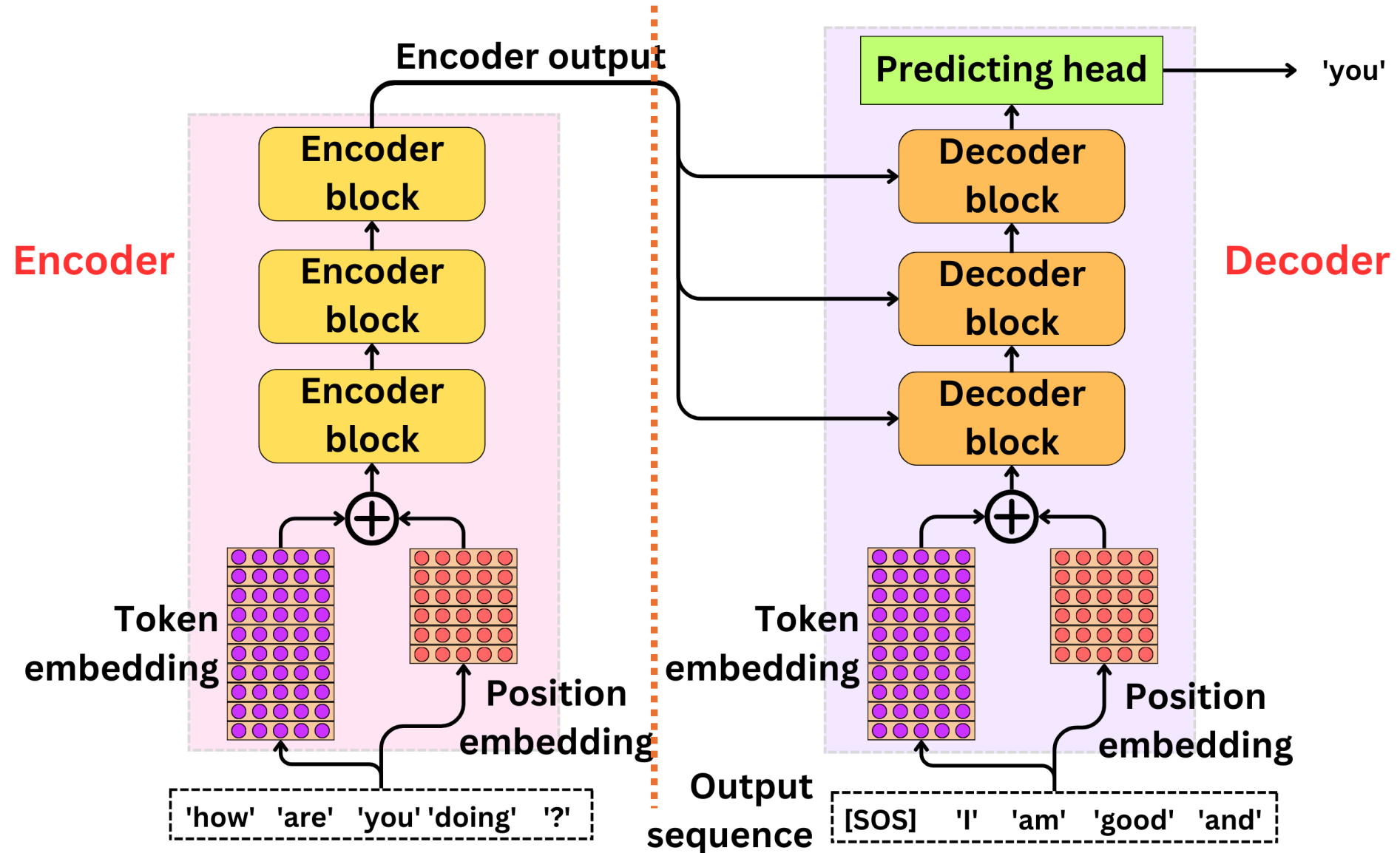


Figure 1: The Transformer - model architecture.

Transformer LM

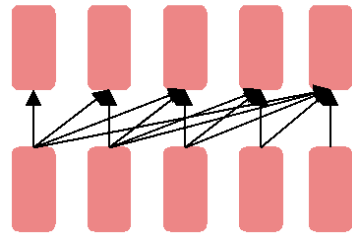


Transformer LM

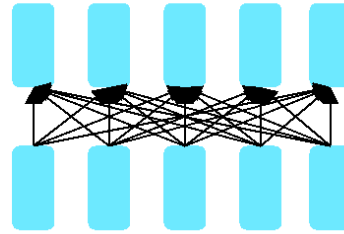


Transformer LM

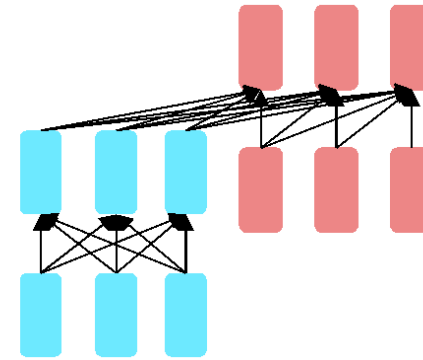
- Three architectures for large language models



Decoders



Encoders



Encoder-decoders

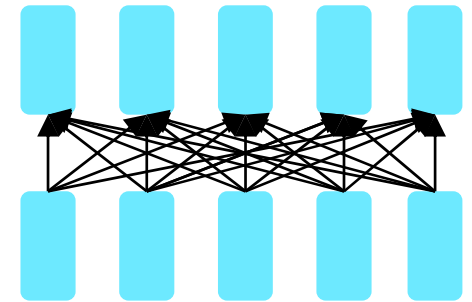
Transformer LM

- Encoders

- Many varieties!

Popular: Masked Language Models (MLMs)

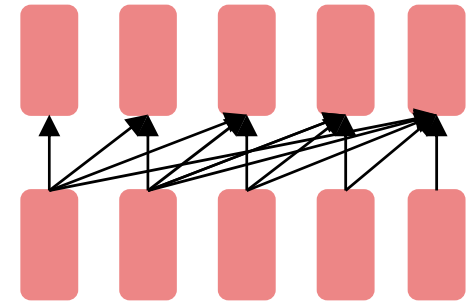
BERT family



- Trained by predicting words from surrounding words on both sides
 - Are usually **finetuned** (trained on supervised data) for classification tasks.

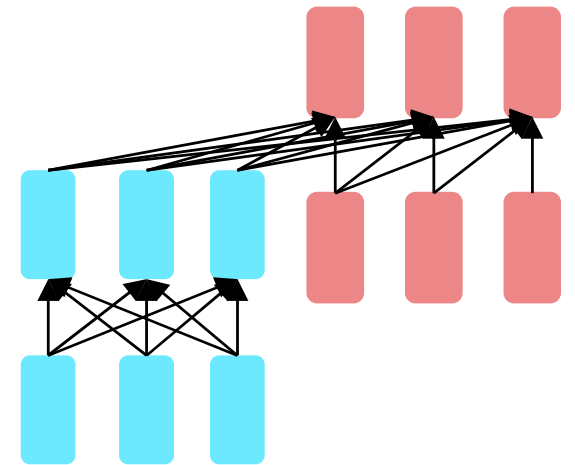
Transformer LM

- Decoder-only models
 - Also called:
 - Causal LLMs
 - Autoregressive LLMs
 - Left-to-right LLMs
- Predict words left to right



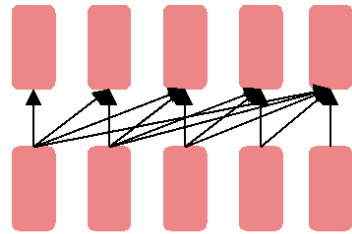
Transformer LM

- Encoder-Decoders
 - Trained to map from one sequence to another
 - Very popular for:
 - machine translation (map from one language to another)
 - speech recognition (map from acoustics to words)



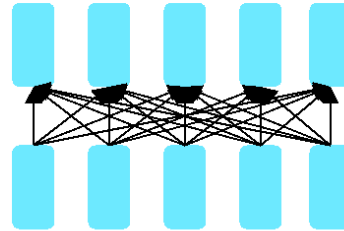
Transformer LM

- Three architectures for large language models



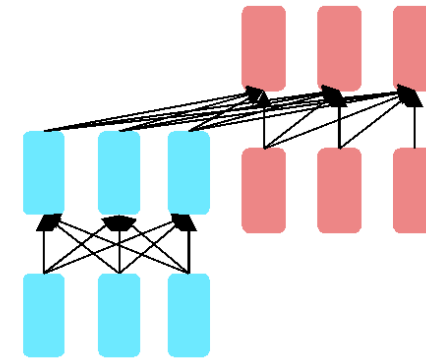
Decoders

GPT, Claude,
Llama
Mixtral



Encoders

BERT family,
HuBERT

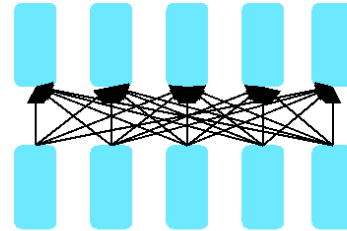


Encoder-decoders

Flan-T5, Whisper

Transformer LM - encoders

- Transformer encoders

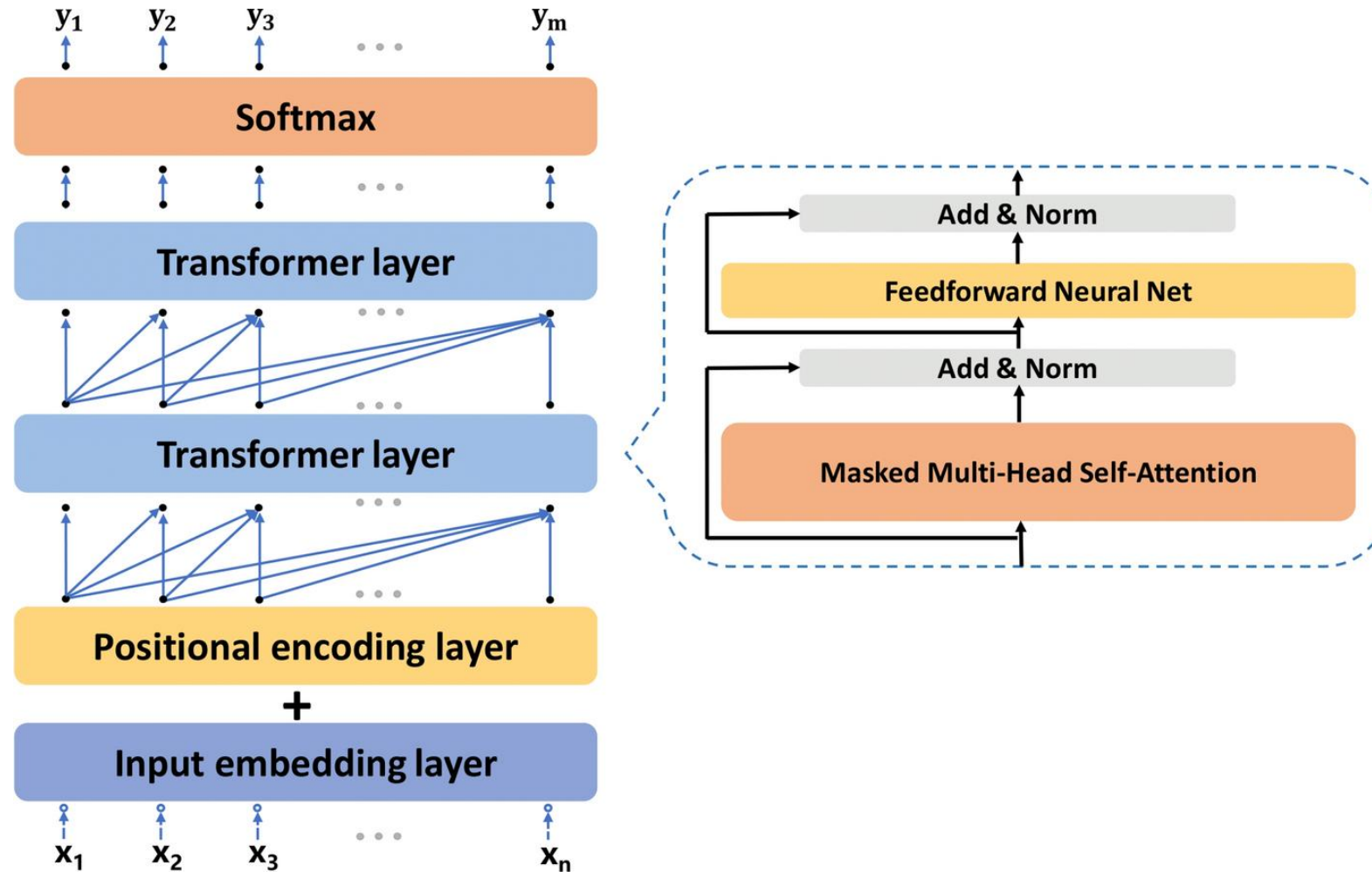


Encoders

BERT family,

HuBERT

Transformer LM - encoders

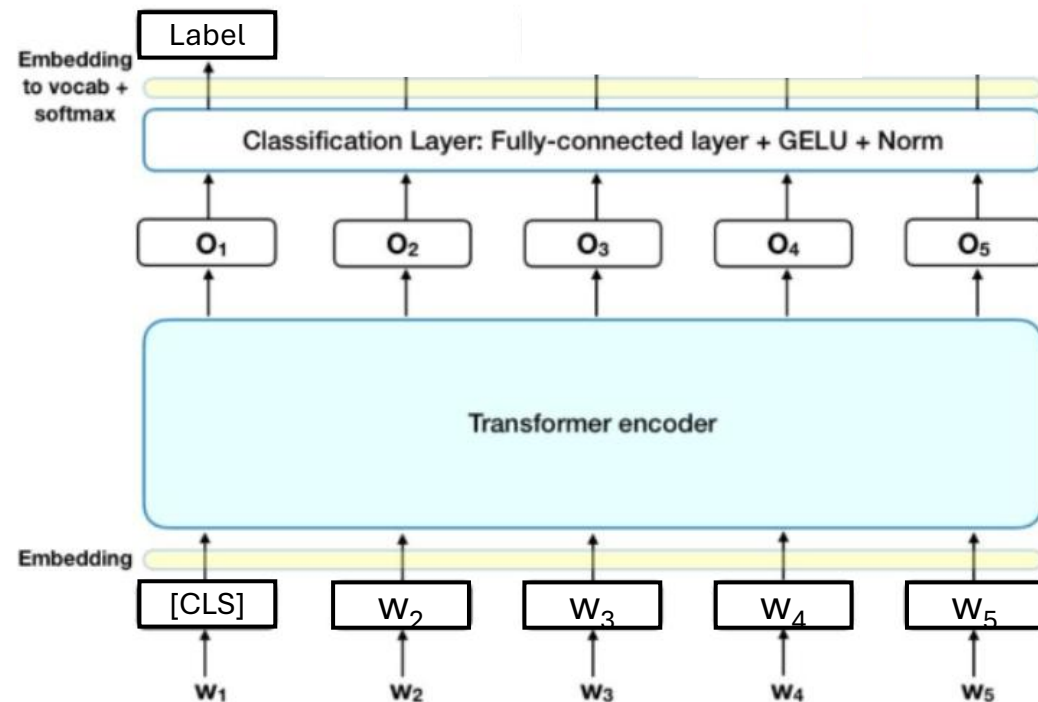


Transformer LM - encoders

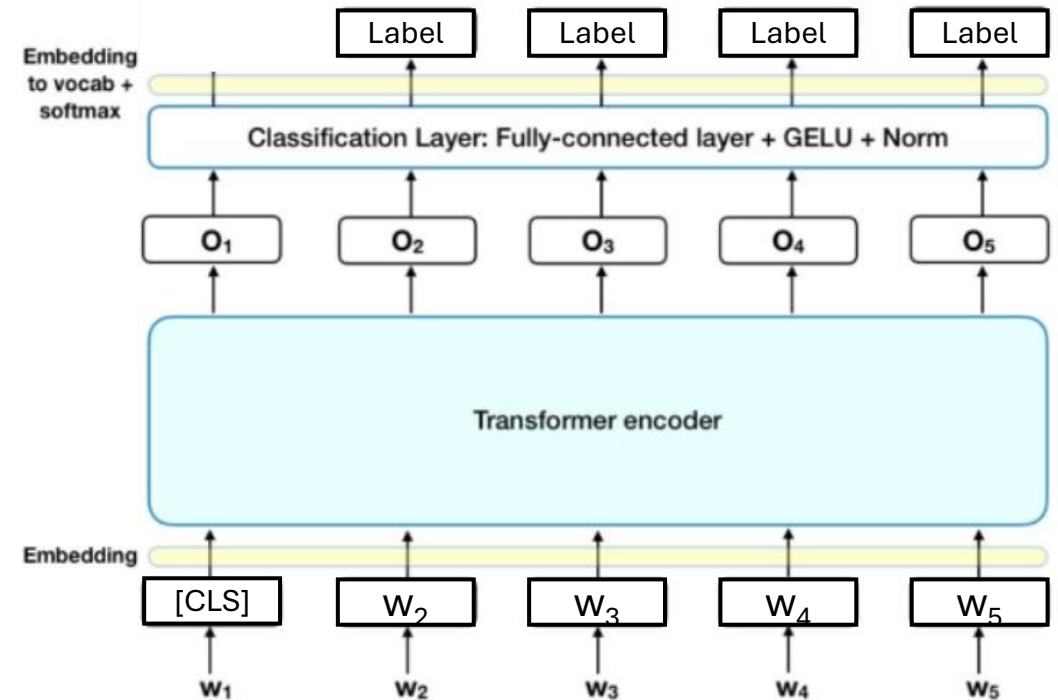
- How to train a BERT model:
 - Masked Language Modeling
 - Permuted Language Modeling
 - Next Sentence Prediction
 - Sentence Order Prediction
 - Replaced Token Detection
- ...

Transformer LM - encoders

- Language Models as Classifiers



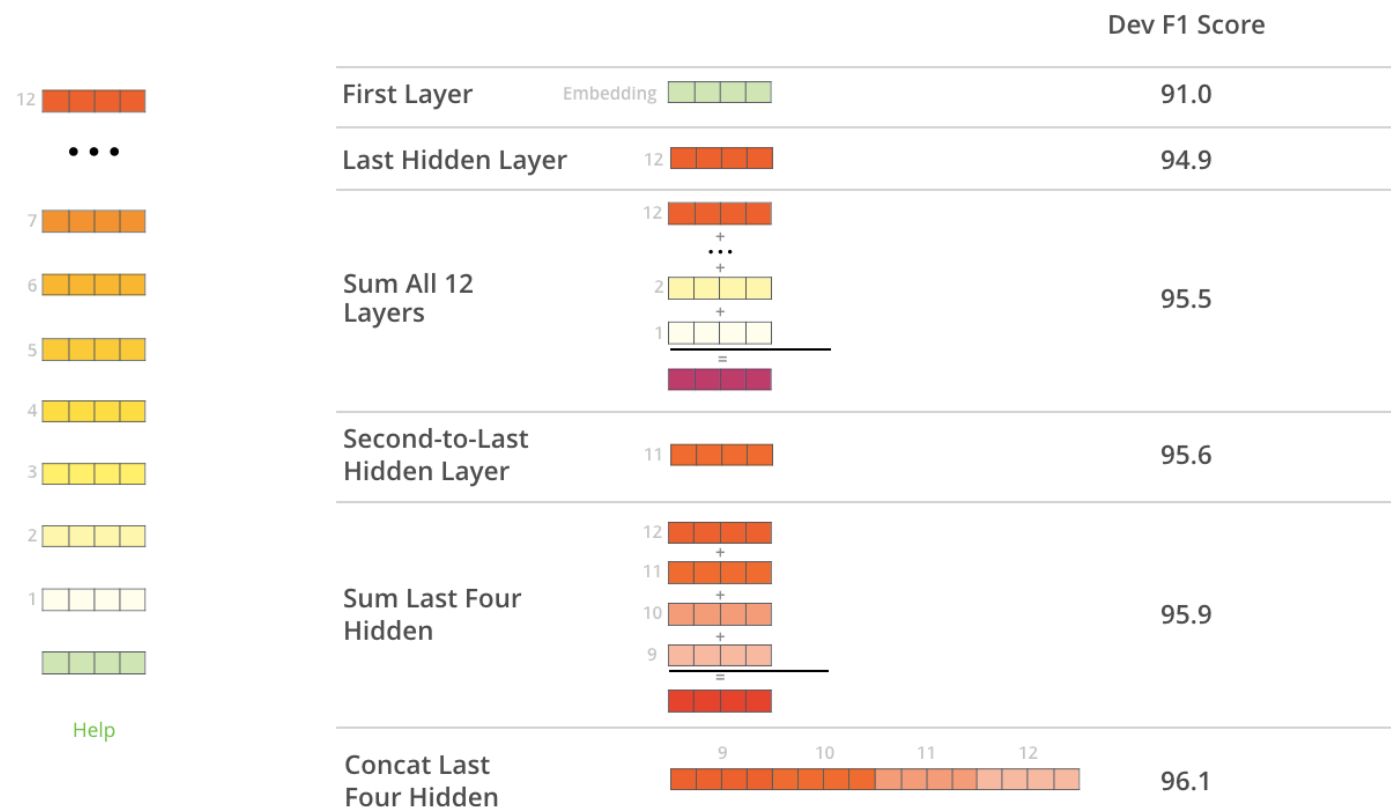
Document/Sentence Classification



Token Classification

Transformer LM - encoders

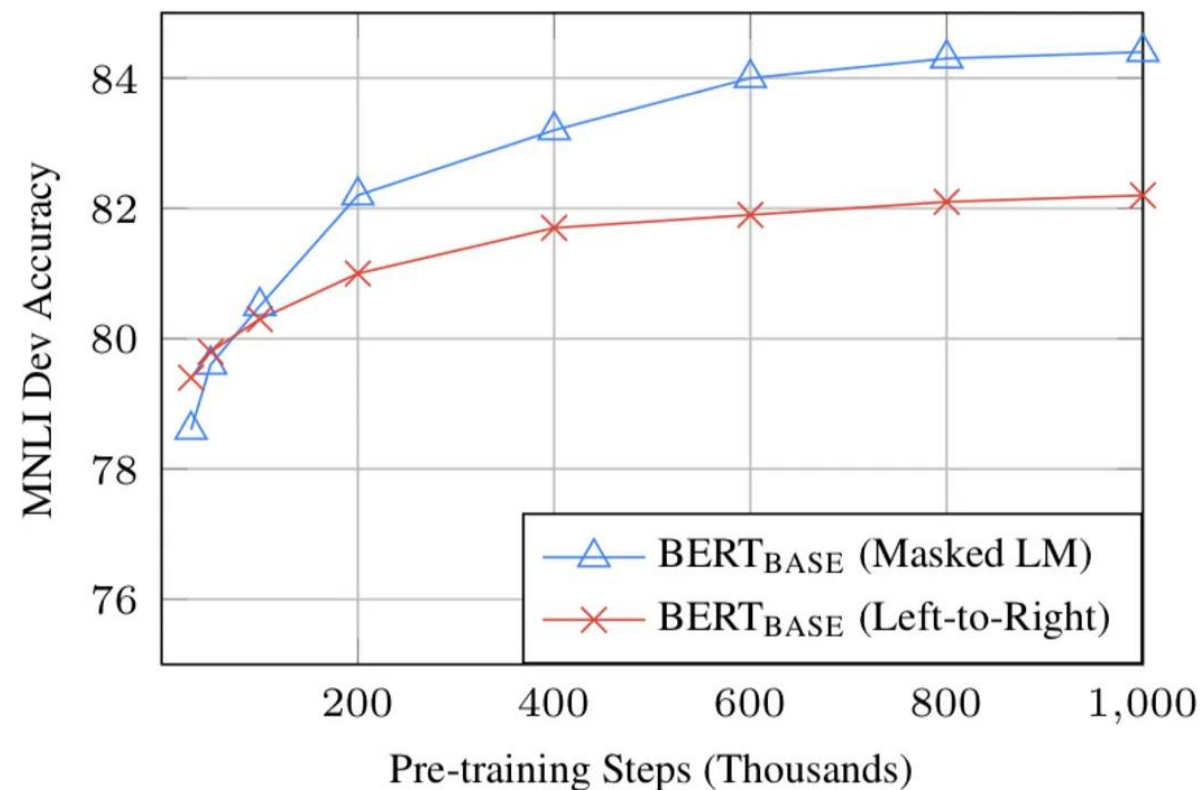
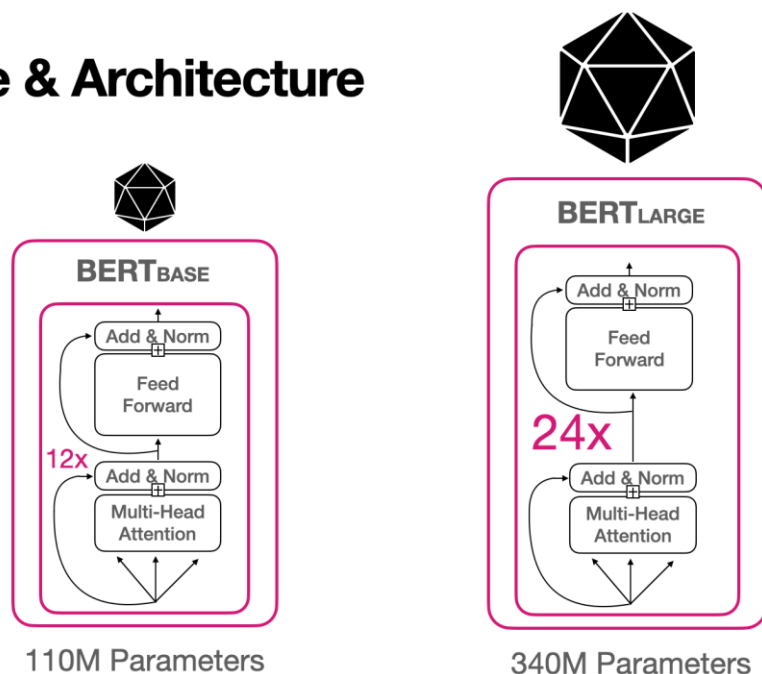
- We can access different model layers



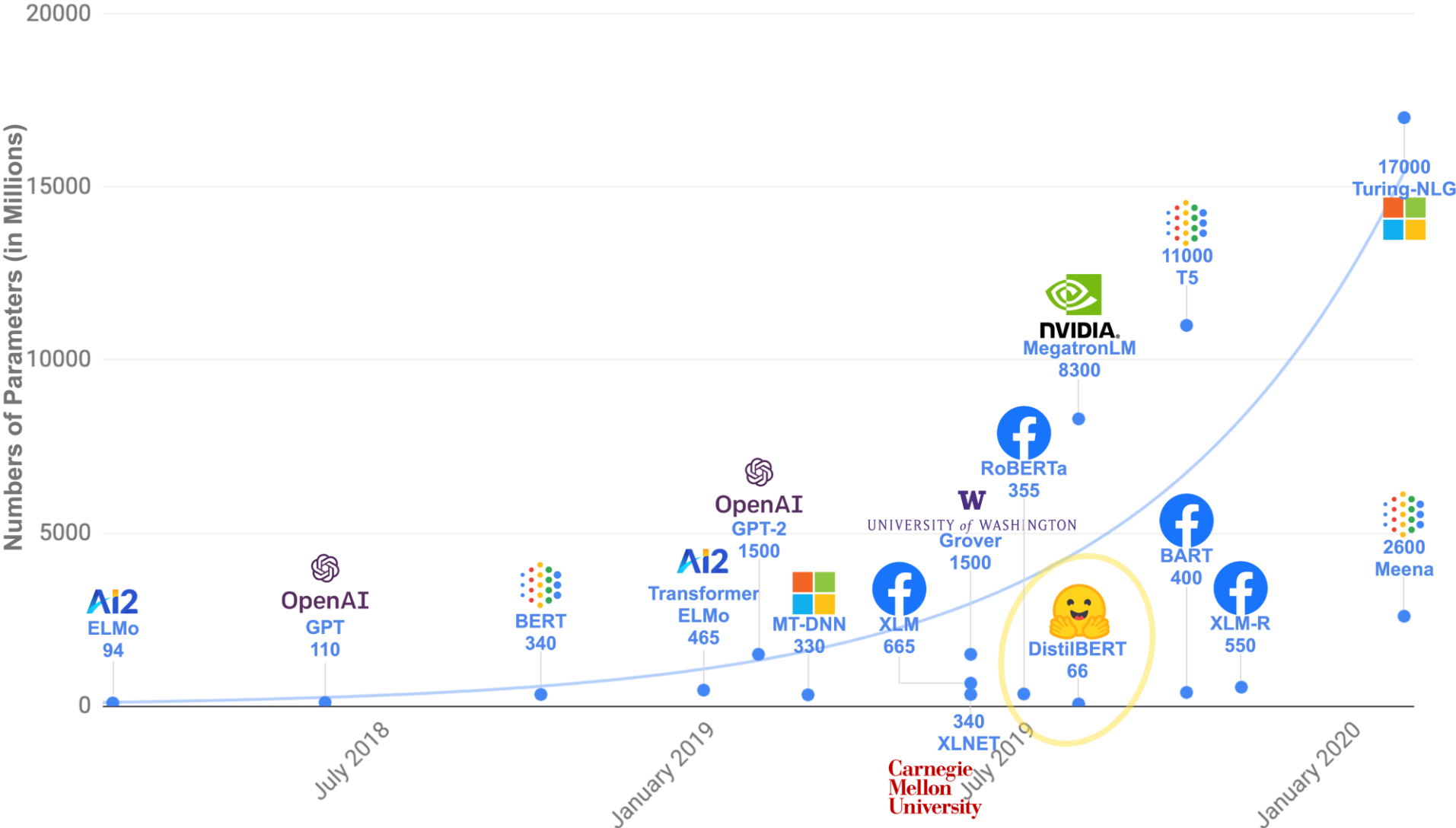
Transformer LM - encoders

- There are different sizes of models

BERT Size & Architecture

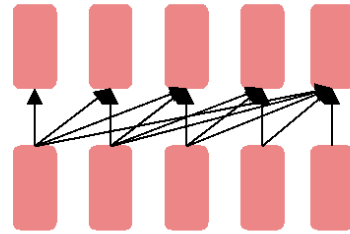


Transformer LM - encoders



Transformer LM - decoders

- Transformer decoders



Decoders

GPT, Claude,

Llama

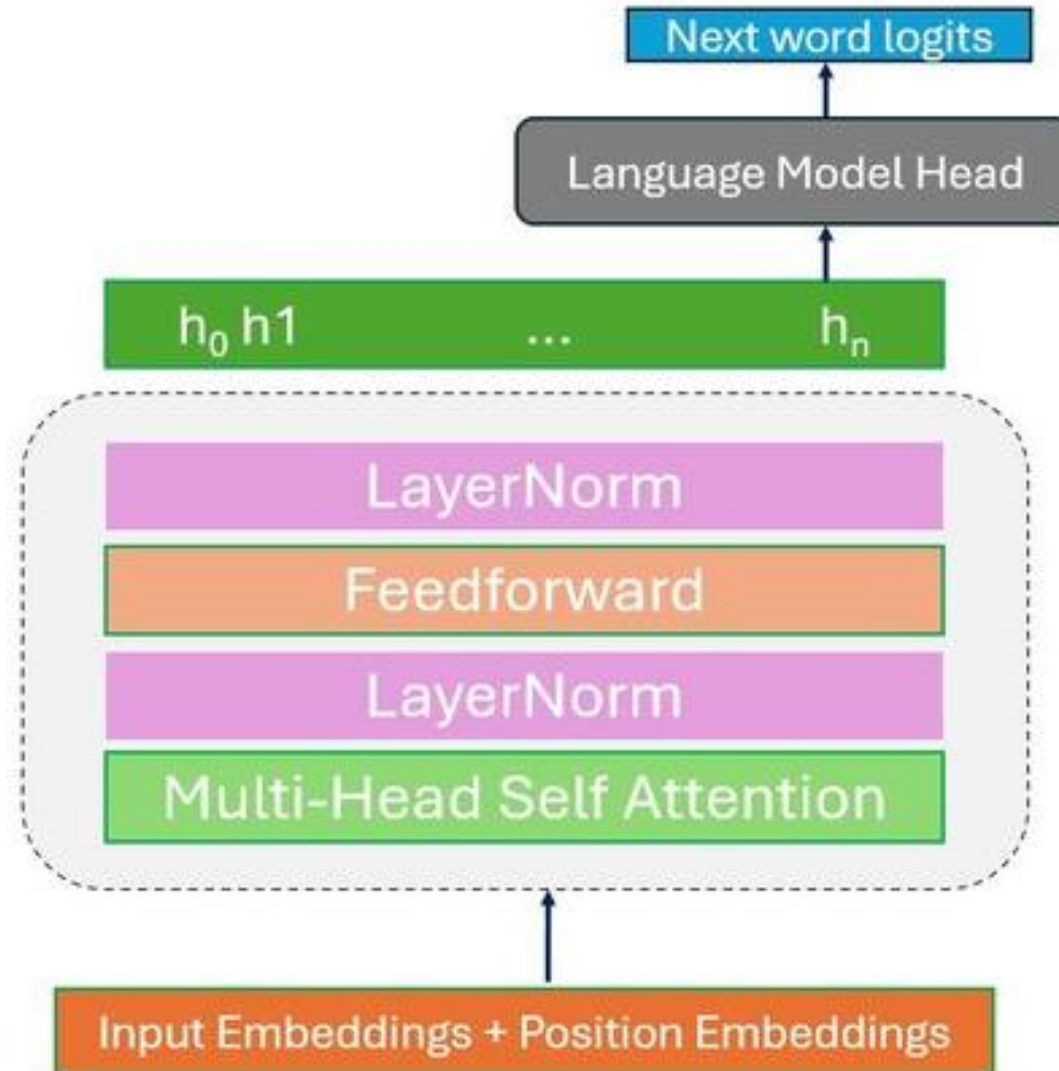
Mixtral

Transformer LM - decoders



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- When will it stop?



Transformer LM - decoders

A SURVEY ON POST-TRAINING OF LARGE LANGUAGE MODELS

We can also specialize

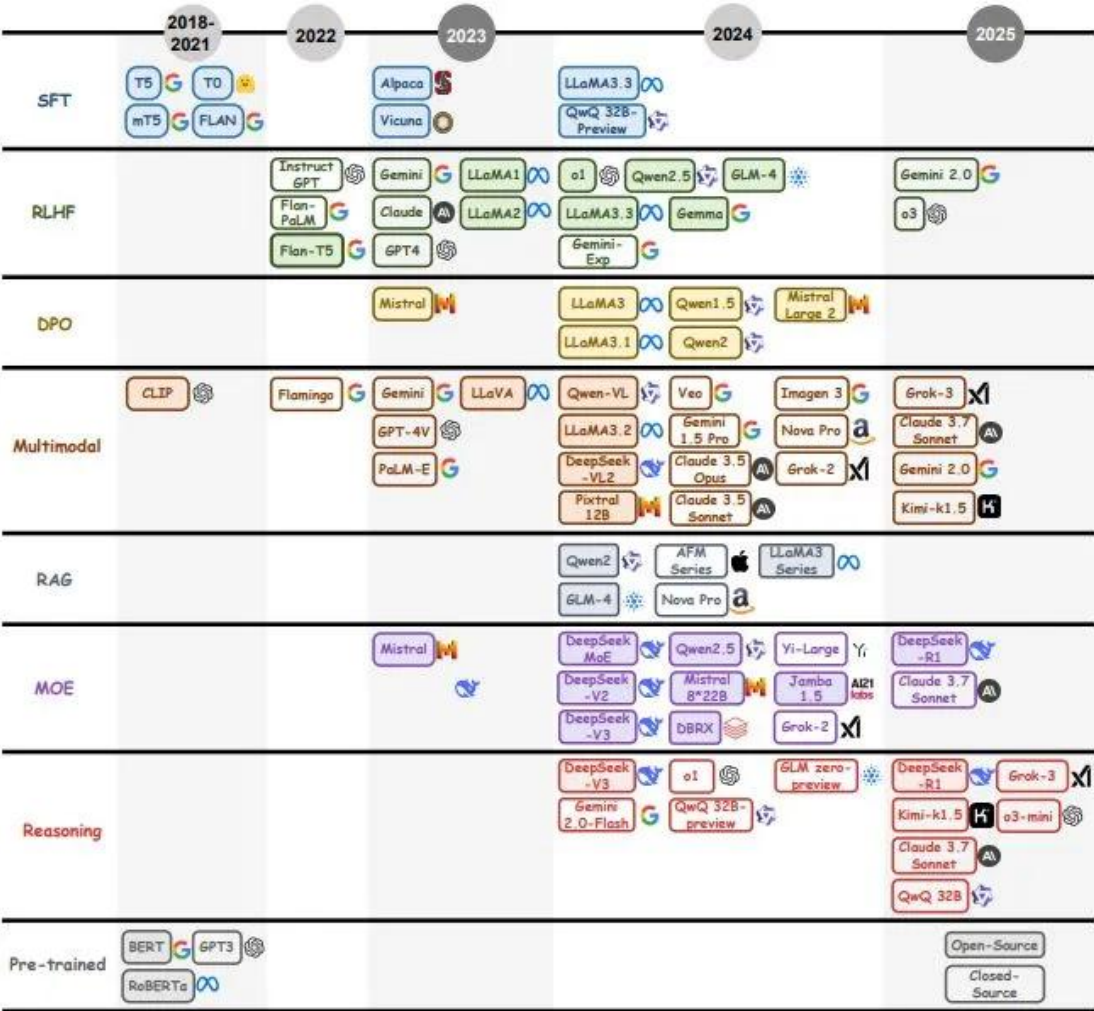


Figure 3: Timeline of post-training technique development for Large Language Models (2018–2025), delineating key milestones in their historical progression.

Transformer LM - decoders

We can interact with the model

Transformer LM - decoders

- Interaction via prompts

Basic Prompt Templates

Summarization	<code>{input} ; tldr;</code>
Translation	<code>{input} ; translate to French:</code>
Sentiment	<code>{input}; Overall, it was</code>
Fine-Grained-Sentiment	<code>{input}; What aspects were important in this review?</code>

Transformer LM - decoders

- Few-shot prompts

Definition: This task is about writing a correct answer for the reading comprehension task. Based on the information provided in a given passage, you should identify the shortest continuous text span from the passage that serves as an answer to the given question. Avoid answers that are incorrect or provides incomplete justification for the question.

Passage: Beyoncé Giselle Knowles-Carter (born September 4, 1981) is an American singer, songwriter, record producer and actress. Born and raised in Houston, Texas, she performed in various singing and dancing competitions as a child, and rose to fame in the late 1990s as lead singer of R&B girl-group Destiny's Child. Managed by her father, Mathew Knowles, the group became one of the world's best-selling girl groups of all time. Their hiatus saw the release of Beyoncé's debut album, *Dangerously in Love* (2003), which established her as a solo artist worldwide, earned five Grammy Awards and featured the Billboard Hot 100 number-one singles "Crazy in Love" and "Baby Boy".

Examples:

Q: In what city and state did Beyoncé grow up?

A: Houston, Texas

Q: What areas did Beyoncé compete in when she was growing up?

A: singing and dancing

Q: When did Beyoncé release *Dangerously in Love*?

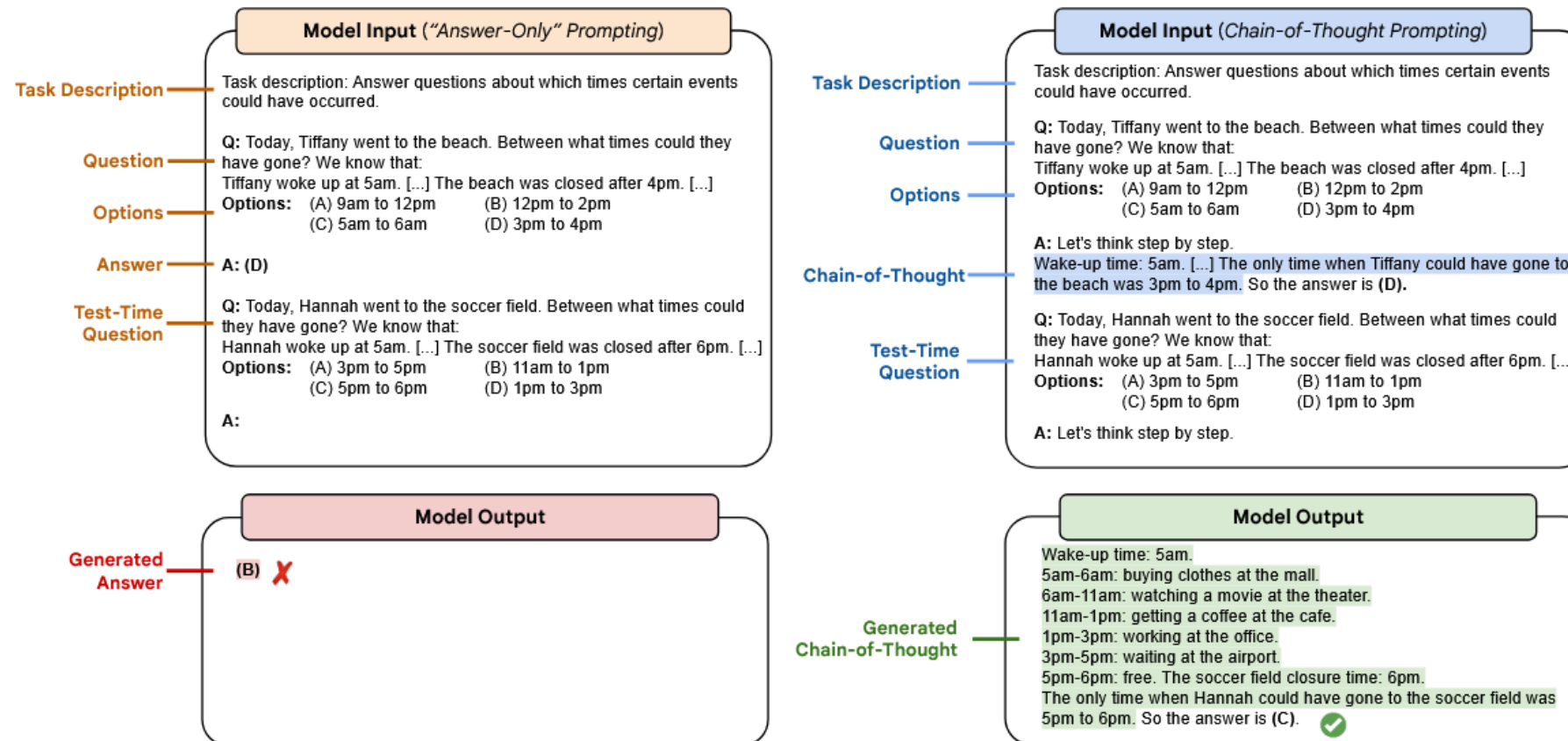
A: 2003

Q: When did Beyoncé start becoming popular?

A:

Transformer LM - decoders

• Chain-of-thought (CoT) prompting



Transformer LM

- Going further:
 - Prompt Engineering vs Fine-Tuning
 - Multimodal Models
 - Open-LLM vs. Closed-LLM

Transformer

- What is the right type of transformer?
 - [menti.com](https://www.menti.com)

Language Models

- Going further:

- Hugging Face



- Ollama



Conclusion

- By the end of this session, students will be able to:
 - Compare n-gram, neural, and transformer language models in terms of context modelling and limitations.
 - Explain how attention and transformer architectures underpin modern large language models and applications.
 - Interpret language-model predictions and representations for text generation, classification, and prompting tasks.